

DIJKSTRA

TECHNICAL ARCHITECTURE DOCUMENT

**AI FOR ACCESSIBILITY AND INCLUSION
AI FOR GOVERNANCE AND PUBLIC GOOD**

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Problem Statement (a contextual view):

Effective governance and **public good delivery** depend on citizens being able to access, understand, and act upon government information. In India, government websites and digital portals are the primary channels for disseminating information about welfare schemes, public services, regulations, and data-driven insights. However, for a large section of the population in the North Eastern Region (NER), these platforms remain inaccessible and difficult to use.

One of the key barriers is **language exclusion**. Most government websites are predominantly available in English or Hindi, with limited or no support for North Eastern languages. As a result, local citizens—especially those in rural, tribal, and low-literacy communities—are unable to directly understand scheme details, eligibility criteria, application processes, or policy information. This forces them to rely on mediators such as middlemen, cyber cafés, or local officials, leading to delays, misinformation, and exclusion from benefits.

The problem is further compounded by the linguistic reality of the North East. Many indigenous languages such as **Khasi, Garo, Mizo, Ao**, and several others are low-resource or primarily oral languages, with limited digital content and, in some cases, no widely standardized or commonly used scripts. Traditional translation-based approaches fail to address this challenge, as they assume text-based literacy and well-defined written forms, which do not reflect the lived realities of many North Eastern communities.

In addition to **local citizens, visitors, professionals, researchers, and tourists from other parts of India** also face accessibility barriers. Government websites and public information portals rarely provide contextual explanations of local rules, governance structures, cultural practices, or region-specific regulations. This lack of cultural and

contextual clarity makes public services and information less accessible to outsiders, reducing inclusivity and hindering social, economic, and tourism-related engagement with the region.

Together, these challenges result in:

- Limited uptake of government schemes
- Dependence on intermediaries
- Reduced transparency and trust in governance
- Exclusion of indigenous language speakers
- Barriers for visitors and non-local users

Ensuring **effective communication through language inclusion and accessibility** is therefore essential for achieving good governance and public good outcomes in the North East. There is a pressing need for a solution that goes beyond basic translation and instead **simplifies, contextualises, and communicates government information in ways that are accessible, culturally aware, and inclusive of both local and non-local users.**

Why Translation Alone Is Not Enough?

Existing language solutions focus on word-for-word translation, which fails to address:

- Complex bureaucratic language
- Data-heavy dashboards and PDFs
- Contextual and cultural understanding
- Role-specific needs (local vs visitor)

What is needed is not translation, but explanation, simplification, and contextual communication.

System Overview

The AI-based Advisory System is a scalable, multilingual, and governance-ready digital solution designed to provide citizens (e.g., farmers) with accurate information on government schemes, agricultural advisories, and policies.

While the solution is intended for integration into official government websites, a neural prototype website has been developed to demonstrate and validate the complete functionality in a controlled environment. The same backend APIs and AI services can be seamlessly integrated into existing government agriculture portals during real-world deployment.

Considerations

The solution has been developed by focusing on the Imphal region of Manipur, keeping the needs of local farmers in mind. The prototype is designed around the working and information structure of the Ministry of Agriculture, so that the advice, schemes, and guidance reflect real government policies and processes.

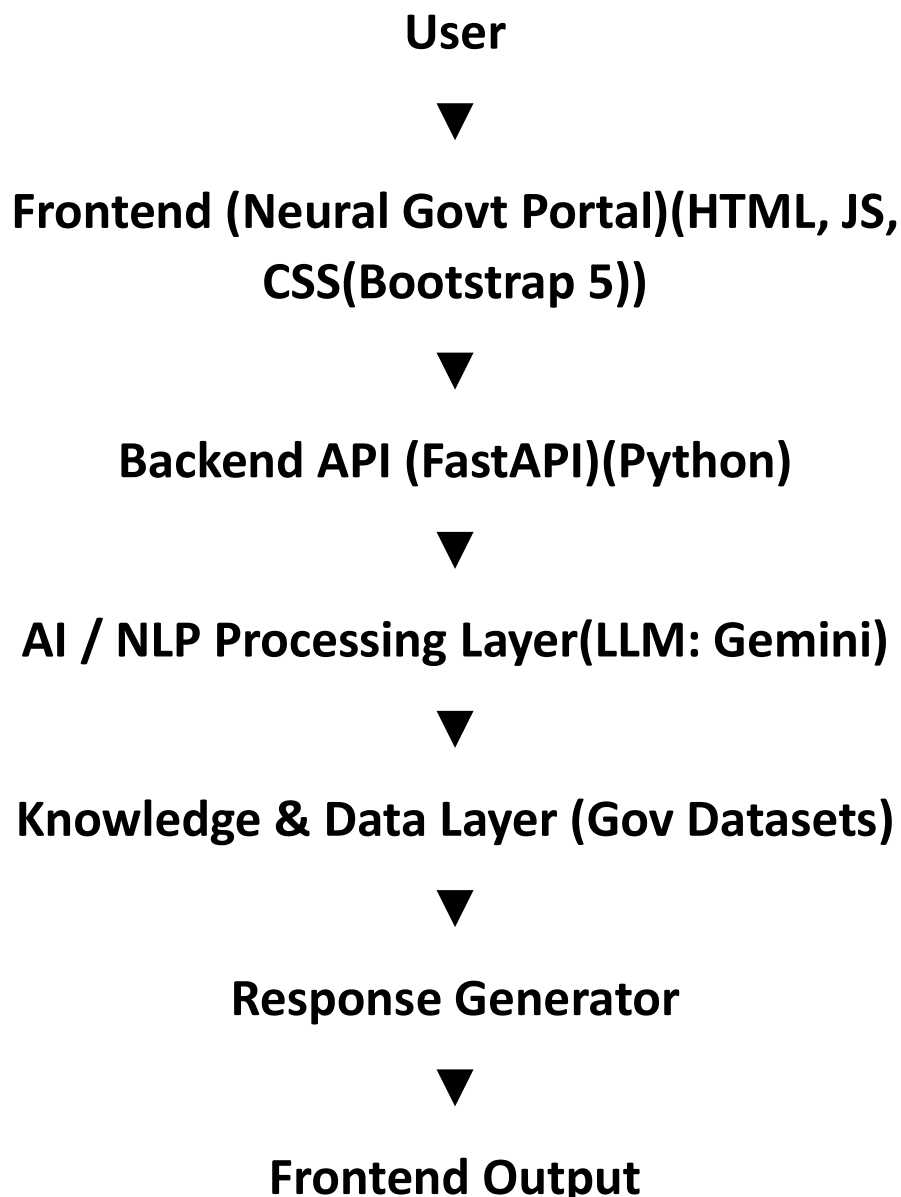
Instead of using an actual government website, the solution has been implemented on a neural, government-style website. This website looks and works like an official Agriculture Department portal. It follows the same layout, navigation, and service flow that users would expect on a real government platform.

This approach allows us to safely test and demonstrate the solution without affecting live government systems. At the same time, the system is built in a way that it can be easily integrated into real government websites in the future.

Although the current prototype focuses on farmers in the Imphal region and supports English and Manipuri languages, the overall design is scalable. The same system can be extended to other districts of Manipur and later to other regions, without major changes to the architecture.

Processing Logic and Workflow Diagram

The following diagram represents the end-to-end technical workflow of the AI-based Farmer Advisory System prototype that we have developed and deployed on a neural (simulated) government website. This same workflow is directly compatible with real government portals.



User Interaction:

Farmers interact with the system via a government-style neural website, using text or voice input. Optional farmer profile details (crop, season, farmer type) enrich the query.

Frontend Layer:

The frontend handles:

- Language selection (English / Manipuri)
- Voice-to-text and text-to-speech
- Clean, accessible UI aligned with government portals

Backend API Layer:

A FastAPI-based backend receives the query, validates inputs, and enriches the request with contextual metadata (region, crop, season).

AI Processing Layer:

The AI model performs:

- **Explanation** (breaking down complex schemes)
- **Simplification** (farmer-friendly language)
- **Contextual reasoning** (profile-aware responses)

Knowledge & Data Layer:

Responses are grounded in **official government documents and advisories**, ensuring accuracy and trustworthiness.

Response Generation & Delivery:

The final advisory is:

- Structured for readability

- Translated into Manipuri (Meitei script) when selected
- Delivered with source references and optional voice output

Feedback Loop:

Users can rate and provide feedback, enabling future system improvement and governance-ready analytics.

Explanation of AI/ML Models Used

. The system uses modern Large Language Models (LLMs) such as Google Gemini and other compatible LLMs.

1. Use of Large Language Models (LLMs)

The solution uses Gemini / LLM-based models for:

- Understanding farmer questions written or spoken in natural language
- Generating clear, structured, and easy-to-understand responses
- Explaining government information in simple terms instead of technical language

These models are already trained on large amounts of general knowledge and language data, which makes them suitable for:

- Question answering
- Explanation and simplification
- Context-aware responses

No sensitive farmer data is used to train the model.

2. Natural Language Understanding (NLU)

The AI model understands farmer queries such as:

- “My crop is destroyed by rain, what can I do?”
- “Which subsidy is available for small farmers?”
- “What schemes are there for paddy in Kharif season?”

The model identifies:

- The intent of the question
- Important context like crop, season, and farmer type

This allows the system to respond correctly even if the question is informal or incomplete.

3. Contextual Reasoning and Personalization

The system enriches the farmer's question using optional inputs like:

- Farmer type (Small / Marginal)
- Crop (Paddy, Maize, Horticulture)
- Season (Kharif / Rabi)

This enriched context is passed to the LLM (Gemini or other LLM), enabling it to provide:

- More relevant and personalized advice
- Practical suggestions instead of generic answers

4. Multilingual Intelligence (English and Manipuri)

The AI logic primarily works in English for accuracy and reasoning.

When the user selects Manipuri, the generated response is translated and displayed in Manipuri (Meitei script).

This ensures:

- Inclusion of local farmers
- Better understanding and trust
- Wider reach in the Imphal region

5. Voice-Enabled AI Interaction

The system supports:

- Voice input using speech-to-text
- Voice output using text-to-speech

This feature is powered by browser-level speech APIs and works alongside the LLM output, making the system accessible even to users who are not comfortable typing.

This makes the system suitable for real government deployment and public service use.

Why This Is NOT an “AI Toy” but a Real Public-Good AI System?

This solution is a production-oriented AI system, not a demonstration or chatbot toy, and is built with real-world governance constraints in mind.

Grounded in Official Government Data

- The AI responses are generated using retrieval-augmented generation (RAG) over official government documents and schemes
- This ensures factual accuracy, traceability, and policy alignment

Designed for Governance & Decision Support

- The system includes feedback collection, usage monitoring, and admin dashboards, enabling government officers to:
 - Understand farmer concerns
 - Improve scheme effectiveness
 - Make data-driven policy decisions

Inclusion of North-Eastern Region (NER)

- The solution prioritizes linguistic and regional inclusion, addressing a key gap in existing digital governance platforms
- By supporting Manipuri language and voice interaction, the system enables access for farmers with limited English literacy

Modular, Secure, and Deployable Architecture

- Built using standard web technologies (REST APIs, FastAPI, vector databases)

- Can be deployed securely within government infrastructure
- Easily auditable and upgradable, making it suitable for long-term public use

Clear Public-Good Impact

This system directly contributes to:

- Digital inclusion
- Farmer empowerment
- Transparent governance
- Equitable access to government schemes

An AI-powered, multilingual farmer advisory system that delivers trusted government guidance in simple language, directly to farmers, enabling inclusive, scalable, and impact-driven digital governance.

ROLES:

- **Pawni Singh– System Architecture & Engineering**
 - Developing the core application codebase and integrating AI components
 - Implementing Retrieval-Augmented Generation (RAG), APIs, and accessibility features
- **Abhinav Tripathi– Data & Model Development**
 - Identification and collection of relevant datasets from Open Government Data (OGD)
 - Designing the data strategy for simplification, regional context, and language inclusion
 - Supporting model tuning and evaluation using public and synthetic datasets
 - Test the system performance, reliability, and end-to-end functionality